CONFIGURING BASIC NETWORK HARDWARETHEORY

1.Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Ans.

Procedure

step.1 First, connect the laptop or computer to the home Wi-Fi network.

step.2 Open Command Prompt by typing cmd in the Windows search bar.

step.3 Type ipconfig and press Enter.

step.4 Find the Default Gateway address. It is usually 192.168.1.1 or 192.168.0.1.

step.5 Open Google Chrome and enter the Default Gateway IP address in the . address bar.

step.6 Press Enter to open the router's admin login page.

step.7 If required, enter the router's username and password to access the admin panel.

step.8 Take a screenshot of the login page by pressing Windows + Shift + S.

step.9 Save the screenshot and attach it to the assignment.

Example

The router's login page may display the router brand name, username field, password field, and Login button.

Example IP Address: 192.168.1.1

Result

The home Wi-Fi router's admin login page was opened using its Default Gateway IP address, and a screenshot was taken for the assignment**.**

2.Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Ans.

Procedure

step.1 First, connect the laptop to the home Wi-Fi network.

step.2 Open the Command Prompt (CMD) on Windows.

step.3 Type ipconfig and press Enter.

step.4 Find the Wi-Fi adapter section in the displayed information.

step.5 Note down the IPv4 Address, Subnet Mask, and Default Gateway.

step.6 Record these details in the assignment.

Observation Table

Sr. No.	Network Details	Example Values
1	IP Address	192.168.1.10
2	Subnet Mask	255.255.255.0
3	Default Gateway	192.168.1.1

Result

The IP address, subnet mask, and default gateway assigned to the laptop through the Wi-Fi network were identified and recorded.

3.Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Ans.

Procedure

1. Connect the laptop to the Wi-Fi network.
2. Open Control Panel and select Network and Sharing Center.
3. Click on Change adapter settings.
4. Right-click on Wi-Fi and select Properties.
5. Select Internet Protocol Version 4 (TCP/IPv4) and click Properties.
6. Select Use the following IP address.
7. Enter the IP address, subnet mask and default gateway according to the router's network settings.

Example settings:

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Preferred DNS Server: 8.8.8.8

8. Click OK to save the settings.
9. Open Chrome and visit a website to check the internet connection.

Result

The laptop's IP address was changed from automatic to static, and the internet connection was checked. The connection worked successfully after applying the correct network settings.

4.In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Ans.

Procedure

1. Connect the laptop to the home Wi-Fi network.
2. Open Chrome and enter the router's IP address, such as 192.168.1.1.
3. Log in to the router's admin panel.
4. Open the **DHCP Settings** or **LAN Settings** section.
5. Find the DHCP starting IP address and ending IP address.
6. Note down the range displayed on the router page.

Observation

Example DHCP Range: 192.168.1.100 – 192.168.1.200

Explanation

If two devices are assigned the same IP address, an IP address conflict can occur. Both devices may face network connection problems, and internet access may become unstable.

Result

The DHCP range was checked in the router's admin panel, and the importance of assigning a unique IP address to each device was understood.